

1. (5 marks) What is Generative AI?

Core Functionality: (3 marks)

Definition of Generative AI: It is a type of AI that can generate new, original content (e.g., text, images, audio, video) based on the data it has been trained on. Key concept: learns patterns and distributions from data to create new, similar outputs. (2 marks)

Examples: LLMs are used for text creation, Imagen are used for image creation etc (1 mark)

Examples of Applications: (2 marks)

Correctly provides two distinct examples. Examples include: code generation, text summarization, content creation, image editing, music composition, drug discovery, design.

2. (4 marks) What is "hallucination" in the context of Generative AI?

Definition: (2 marks) A "hallucination" in GenAI refers to the generation of outputs that are factually incorrect, nonsensical, or inconsistent with the input prompt or the model's training data. (1 mark)

Underlying Causes & Consequences: (2 marks) This is often due to limitations in the training data, biases in the data, or the model's tendency to overfit or extrapolate beyond its knowledge base. Consequences could include: Spread of misinformation, reputational damage, biased outcomes.

3. (4 marks) What does RAG stand for?

Definition: (1 marks) Retrieval-Augmented Generation

Description and Use Case: (3 marks)

RAG is an architecture where a GenAI model is enhanced with an external knowledge source. Given an input query, the system first retrieves relevant information from a database or knowledge base, then uses that retrieved information to condition the generation process, thus improving accuracy and reducing hallucinations.

Good example of use case: Answering questions about a company's internal documents, summarizing complex research papers by retrieving relevant information.

4. (3 marks) How does Streamlit help in building a frontend interface quickly?

Reason 1: (1.5 mark)

Streamlit allows developers to build interactive UIs with minimal code. It abstracts away the complexities of traditional web development. It has built-in functions like `st.button`, `st.select` etc to help in building quickly.

Reason 2: (1.5 mark)

It is very fast in re-rendering and automatically updates the UI when code changes are made, making it easy to prototype and iterate on ML applications. It can also automatically update the app when the python file is saved with the changes

5. (4 marks) Discuss any one open-source Large Language Model (LLM).

Model Name: (0.5 mark) Example: Llama 2, Mistral

Key Features: (2 mark) Examples:

Llama 2: Open-source, different parameter sizes (7B, 13B, 70B), strong performance on various benchmarks, designed for research and commercial use.

Mistral: One of the LLMs with smallest size but yet competitive to other models.

Intended Purpose: (1.5 marks) Examples:

Llama 2: Research, education, and commercial applications, text generation, language understanding, dialogue.

Mistral: Similar to Llama2 but with a different training technique